



1. RESEARCH QUESTION



Is the proportion of current cigarette use different between students who felt sad or hopeless and those who did not?



2. DATA & VARIABLES



Dataset: YRBS 2007

Total Sample Size

13,174

Group Variable

- SadOrHopeless (Binary)
 - 1 = Felt sad or hopeless (Yes)
 - 0 = Did not feel sad or hopeless (No)

Response Variable

- CurrentCigaretteUse (Binary)
 - 1 = Current smoker (codes 2-7)
 - 0 = Non-smoker (code 1)



3. METHOD



Two-Proportion Z-Test

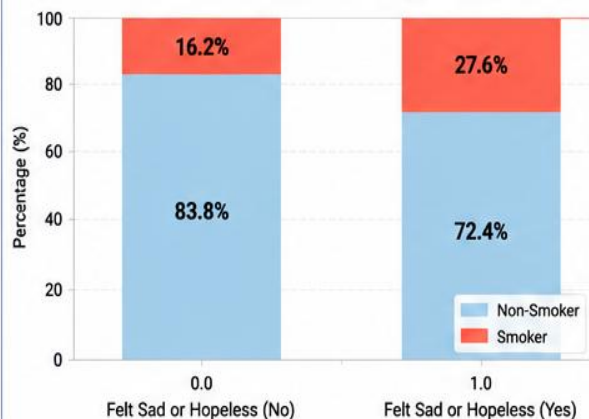
$$H_0: p_1 - p_0 = 0$$

$$H_a: p_1 - p_0 \neq 0$$

Significance Level: $\alpha = 0.05$ 

4. EDA HIGHLIGHT: SMOKING STATUS BY SADNESS

100% Stacked Bar Chart of Smoking Status by Sadness



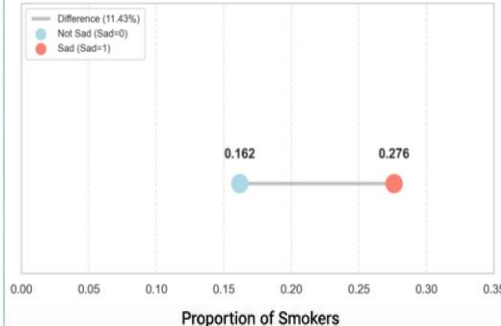
Interpretation

Students who felt sad or hopeless have a much higher proportion of current smokers (27.6%) compared to those who did not (16.2%).



5. DIFFERENCE IN PROPORTIONS

Dumbbell Plot: Gap in Smoking Proportions



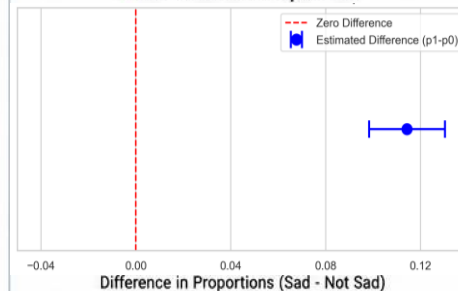
Interpretation

The proportion of smokers among students who felt sad or hopeless is 11.43 percentage points higher.



6. STATISTICAL INFERENCE

95% Confidence Interval for the Difference in Proportions



Interpretation

The 95% confidence interval (0.1076, 0.1209) does not include zero, indicating a statistically significant difference.



7. KEY RESULTS

Smoking Rate
(Sad = 1)

27.61%

Smoking Rate
(Sad = 0)

16.18%

Difference
($p_1 - p_0$)

11.43%



Z Statistic

15.05



P-value

3.267e-51

95% Confidence
Interval(0.1076,
0.1209)

8. CONCLUSION



The analysis demonstrates that adolescent students experiencing sadness or hopelessness exhibit an 11.43% higher smoking prevalence compared to those without such emotional distress. This confirms a strong statistically significant association, suggesting that negative emotional health may be a critical risk factor for adolescent cigarette use.

